Assignment–02: Comparative Study and Practical Analysis of Generative AI Models

Group Assignment (CO3 and CO4)

Name	PRN	Role in report
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GitHub Repository: https://github.com/Rushikeshaya/Generative-AI/tree/main/Assignment%20No.%202%3A%20(AI-Assisted)%20Comparative%20Study%20and%20Implementation%20of%20Generative%20AI%20Models

1. Introduction

Generative AI models learn the structure of image data so they can either reconstruct an input or create new images. This report compares four models: the Autoencoder (AE) and Variational Autoencoder (VAE) for image reconstruction (CO3), and the Deep Convolutional GAN (DCGAN) and a pretrained Diffusion Model, Stable Diffusion, for image generation (CO4). It also analyses the outputs obtained by running ready-made notebooks in Google Colab. No model was trained from scratch for the diffusion part.

2. Task 1: Comparative Study of the Four Models

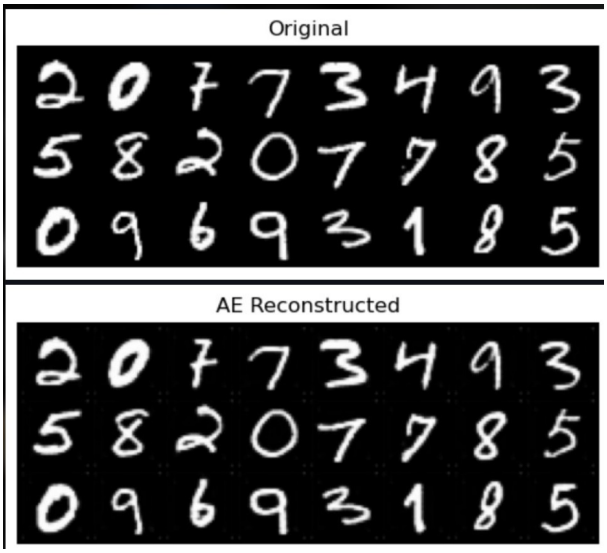
Criteria	Autoencoder (AE)	Variational Autoencoder (VAE)	DCGAN	Pretrained Diffusion (Stable Diffusion)
Working principle	Compresses the input into a small latent vector and rebuilds it. Trained to minimise reconstruction error (MSE).	Encodes the input as a distribution (mean and variance), samples a latent vector, then decodes it. Loss = reconstruction + KL divergence.	Generator makes fake images from random noise; discriminator tells real from fake. Both improve through a min-max game.	Learns to remove noise step by step. Generation starts from pure noise and is denoised over many steps, guided by a text prompt.
Architecture and major components	Encoder, bottleneck (latent code), decoder.	Encoder (mean and log-variance), reparameterisation trick, decoder.	Generator (transposed convolutions, batch norm, ReLU) and discriminator (strided convolutions, LeakyReLU).	CLIP text encoder, U-Net denoiser in latent space, VAE encoder/decoder, noise scheduler.
Advantages	Simple, fast to train. Good for compression, denoising and feature learning.	Smooth, continuous latent space. Can generate new samples. Stable training.	Sharper images than AE/VAE. Fast generation in a single forward pass.	Very high quality and realism. Text-to-image control. Stable training. Diverse outputs.
Limitations	Irregular latent space, so poor at generating new images. Loses fine detail.	Blurry outputs because of the KL term and averaged likelihood.	Unstable training, mode collapse, sensitive to hyperparameters. Small image sizes.	Slow (many steps). Needs GPU and large memory. May distort hands or text and can reflect dataset bias.
Applications	Denoising, dimensionality reduction, anomaly detection, compression.	Data generation, latent interpolation, anomaly detection, representation learning.	Face and image synthesis, data augmentation, style transfer, super-resolution.	Text-to-image art, design, inpainting, image editing, content creation.
Output differences	Close to the input, slightly blurred.	Smoother and blurrier than AE, but more meaningful latent space.	New images that are sharp but may contain artefacts or repeating patterns.	Detailed, realistic images following the prompt; occasional anatomy or text errors.

Summary: AE and VAE are reconstruction-focused. The VAE adds a probabilistic latent space that supports generation, but costs sharpness. DCGAN improves sharpness through adversarial training but is harder to train. Diffusion models give the best realism and prompt control at the highest compute cost.

3. Task 2: Practical Implementation and Output Analysis

Setup: Google Colab with GPU runtime, ready-made notebooks for AE, VAE and DCGAN (PyTorch), and the Hugging Face diffusers library for Stable Diffusion. Dataset used for AE, VAE and DCGAN: MNIST (28x28 grayscale handwritten digits). Stable Diffusion prompt: "_____".

Activity A: Image Reconstruction (CO3)

Dataset: MNIST training samples	Original (top) vs Autoencoder reconstructed (bottom)
	

Aspect	Autoencoder	VAE
Blurring	Very little; strokes stay sharp with only slight softening at the edges.	Noticeably smoother and blurrier overall.
Distortion	None visible; every digit keeps its shape.	Slight shape changes; a digit may look like an averaged version of its class.
Loss of detail	Minor; faint dotted noise appears on the black background.	More fine detail lost because of the KL regularisation.

Analysis: In our run the AE reconstructions are almost identical to the original MNIST digits, because the AE only minimises reconstruction error and its latent code can store input-specific detail. The VAE is forced to keep its latent distribution close to a standard normal, which smooths the latent space and makes sampling possible, but removes some detail and produces blurrier reconstructions. So AE gives better fidelity, while VAE gives a better-structured latent space.

Activity B: Image Generation (CO4)

DCGAN generated digits	Stable Diffusion generated image
	

Aspect	DCGAN	Stable Diffusion
Image quality	28x28 grayscale digits; most are readable but strokes are uneven.	High resolution (512x512), sharp and clean.

Aspect	DCGAN	Stable Diffusion
Realism	Looks like handwriting, but many digits are ambiguous or malformed.	Very high; natural fur texture, lighting and water reflections.
Details	Broken or merged strokes and noisy edges.	Rich detail in fur, flowers, rocks and skyline; slightly over-saturated look.
Limitations	Some samples are not valid digits; no control over which digit appears.	Needs a GPU and many denoising steps; scene looks idealised and slightly artificial.

Analysis: Our DCGAN generates digits in one forward pass and is therefore fast, but its output is small and many samples have broken or ambiguous strokes, which is typical of adversarial training. Stable Diffusion needs dozens of denoising steps and a GPU, yet produced a far more realistic and detailed image from a text prompt. For realism and control, diffusion is clearly superior; for speed and lightweight deployment, DCGAN is still useful.

Note: *AE, DCGAN and Stable Diffusion observations are taken from our screenshots. The VAE column describes standard VAE behaviour.*

4. Task 3: AI Usage Documentation and Group Reflection

AI tools used: Claude (drafting, concept explanation); _____ (add ChatGPT / Gemini / Copilot if used).

Important prompts used:

1. Compare Autoencoder, VAE, DCGAN and Stable Diffusion in a table on working principle, architecture, advantages, limitations, applications and output differences.
2. Explain why VAE reconstructions are blurrier than Autoencoder reconstructions.
3. Stable Diffusion image prompt: "_____".

Verification process: AI-generated content was cross-checked against the original papers (Kingma and Welling 2013; Radford et al. 2015; Rombach et al. 2022) and the PyTorch and Hugging Face documentation. Practical observations were checked against our own notebook outputs.

Key learning outcomes:

- Reconstruction and generation models solve different problems; the choice depends on fidelity, sharpness or diversity.
- The VAE's KL term trades sharpness for a smooth, usable latent space.
- Adversarial training gives sharp outputs but is unstable; diffusion models give the best quality at higher compute cost.

Contribution of each member:

- **Soham Sabane (202401110070):** Ran the Autoencoder and VAE notebooks, captured screenshots and wrote the Activity A analysis.
- **Rushikesh Hande (202401110056):** Ran the DCGAN and Stable Diffusion notebooks, captured screenshots and wrote the Activity B analysis.
- **Yadnesh Tayade (202401110057):** Prepared the comparative table, AI usage documentation, PPT and presentation.

References

[1] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," 2013. [2] A. Radford, L. Metz and S. Chintala, "Unsupervised Representation Learning with Deep Convolutional GANs," 2015. [3] R. Rombach et al., "High-Resolution Image Synthesis with Latent Diffusion Models," CVPR 2022. [4] PyTorch DCGAN tutorial and Hugging Face Diffusers documentation.